

IBM z17

Advanced AI where
it matters most

Seller enablement presentation

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All client examples described are presented as illustrations of how those clients have used IBM products and the results they may have achieved. Actual environmental costs and performance characteristics may vary by client.

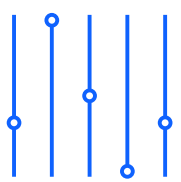
The IBM Z mainframe

IBM Z (the Z stands for “zero downtime”) is a cornerstone of enterprise IT and has served as the backbone of the global economy for decades

A resilient technology foundation is essential for organizations that want to better prepare for the future



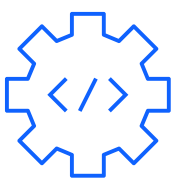
Unparalleled speed,
scalability, and security
quietly running the world’s
critical workloads



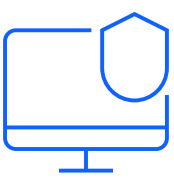
Fine-tuned and
highly-performant
processing



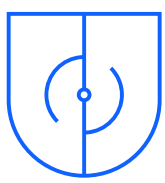
Highly scalable



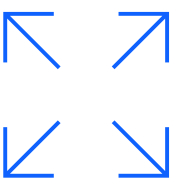
Concurrent
upgrades
and repairs



Encryption
everywhere



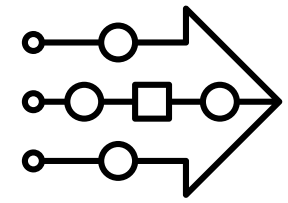
Quantum-safe
protection



Extreme utilization

IBM z17 makes more possible

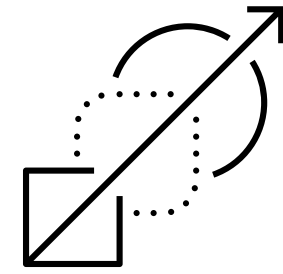
Enhance your hybrid cloud with advanced AI where it matters most



Fueling innovation and growth with AI

- Deliver business growth with deep insights
- Unleash the power of generative AI on data*
- Drive greater impact from your most important data

AI for business



Automating and transforming for efficiency

- Accelerate mainframe application modernization with generative AI*
- Streamline IT Operations with AI insight
- Boost productivity and automate tasks



Securing the most important data

- Proactively identify and mitigate cyber threats
- Quantum-safe protection
- Simplify compliance and improve productivity

Optimizing and Securing your Infrastructure with **AI**

* Upon IBM Spyre AI Accelerator availability

IBM z17

Value of currency



Price performance

- **11%** Single thread performance faster response times, improving batch window resiliency
- **15-20%** more capacity within existing footprint to growth within the same frame
- **60%** more system memory capacity to improve online application response time
- Improved software pricing per MIPS
- Next generation of Tailored Fit Pricing for AI and OpenShift zCX workloads

Resiliency

- Accelerate your quantum-safe journey with simplified cryptographic inventory and updated quantum-safe APIs based on NIST PQC standards
- Improve productivity and automate tasks with generative AI while keeping your data secure and in place
- Flex Capacity offering, exclusive to z17, for infrastructure testing

Energy efficiency

- **70%** power reduction on I/O management
- A typical IBM z17 system can reduce weight by **13%** and is estimated to reduce system power consumption by approximately **19%** compared to a similarly configured IBM z16
- A max config IBM z17 system can reduce weight by **26%** and is estimated to reduce system power consumption by approximately **27%** while using **25%** less floor space when compared to a similarly configured IBM z16

Design thinking

IBM z17 co-created with clients

81

enterprises

*Compared to
70 at z16 GA*

IBM Z Sponsor user program
IBM Z Design Council
GM Advisory Council
Cross section of users
(geos, industries, size)
1 on 1 Interviews and Surveys
Hill value validation
Prototype evaluations

2000+

interaction
hours

*Compared to
1107 at z16 GA*

Nearly 2x hours versus IBM z16
from developers, architects, to execs

Driven by early engagements across
IBM Z and LinuxONE

Innovations through cross-team
alignment from product
management, design, marketing,
development, sales
and support

36

user
personas

Application Architect
Application Developer
Application Owner
Auditor
Chief Data Officer
Chief Privacy Officer
Chief Sustainability Officer
Cloud Architect
CTO, CIO, CFO, CISO / CIRO
Data Architect
Dev Ops Engineer
Director of Mainframe Datacenter
Evidence Provider
Infrastructure Architect
Line of Business Owner
Security Architect
System Admin
VP of AI / Analytics
VP of Application Development
VP of Infrastructure

IBM z17:
Powering the world's
economy in the AI+ era

Reasons to upgrade



Upgrade to IBM z17 to make more possible

Advanced AI where it matters most to fuel business growth, secure your most important data on an industry–leading resilient system, and enable automation to transform for operational efficiency

Capability	Proof points and use cases	IBM z17	IBM z16	IBMz15
In transaction AI at speed and scale, performance	- Telum II supports encoder language models (LLMs): 8 AI processors allows routing of inference requests within the same drawer to increase inference throughput by up to 7.5x as compared to IBM z16. ¹ - Up to 450 billion inference operations per day with 1ms response time.² (300 billion on IBM z16) 11% single thread performance improvement³ , 40% more cache⁴	5nm , 5.5GHz Telum II 7.5x AI throughput 5M inferences/second	7nm, 5.2GHz Telum 3M inferences/ second	14nm
Generative AI and multi-model AI	IBM Spyre AI Accelerator (Avail. Q42025) enables generative AI on-premises and works ; up to 48 per system. With Telum II to support encoder and decoder large language models (LLMs) and multi-model AI. Process up to 450 billion inference operations per day using multiple AI models⁵	X IBM Spyre AI Accelerator		
AI-Powered security	- Sensitive Data Tagging for IBM z/OS (Avail. Q42025) leverages AI acceleration of Telum II chip and tuned odels to routinely scan, discover, classify, and tag sensitive data - IBM Threat Detection for z/OS uses AI to identify anomalies in data access that might indicate a cyberattack; can support an anti-malware solution as required by regulations, such as DORA and PCI DSS 4.0	X X	 X	
Quantum-safe security	- Utilize powerful post–quantum algorithms endorsed by NIST & recommended by CNSA 2.0, to help protect against the emerging quantum threat landscape; IBM Crypto Express8s utilizes NIST standardized algorithms. - Discover and classify cryptography with IBM Z Crypto Inventory and Discovery	X+ X	X X	 X
Scalability	15-20% more capacity⁶ · 60% more memory than IBM z16 - Capable of processing up to 35 billion encrypted OLTP transactions per day⁷ - Can deploy up to 3,000,000 NGINX containers running Red Hat Enterprise Linux with KVM⁸	208 Cores 64 TB RAM 35B encrypted transactions	200 Cores 40 TB RAM 25B encrypted transactions	190 Cores 40 TB RAM
Industry-leading resiliency and availability	- Industry-leading 99.999999% availability equiv. to 315 mis of downtime per year⁹ - IBM Z Flexible Capacity for Cyber Resiliency optimizes resources between systems in different data centers within minutes for disaster recovery, compliance and testing	8 9’s availability X	8 9’s availability X	7 9’s availability
Workload Classification Pricing	Deploy new workloads with consumption based pricing , and improved price-to-value. Minimizes up-front investments with no long-term hardware commitments allowing dynamic capacity scale	X		
Energy efficiency, optimization	- Move AI-infused OLTP workloads from compared x86 can save up to 83% of power consumption¹⁰ - Move cloud-native, containerized workloads from compared x86 can save up to 65% of power consumption ¹¹ - Save up to 44% on TCO over 5 years by moving cloud-native, containerized workloads from compared x86 ¹² 17% less power consumption than IBM z16 ¹³	19% less power consumption		

Hardware and system structure improvements

Comparison of IBM z16 to IBM z17

	z16	z17	z17 differentiation
Processor Chip	7nm technology 5.2G Hz 8 cores per chip 18.8 Miles of wire per chip 22.5B transistors	5nm technology 5.5 GHz 8 cores per chip 24.1 Miles of wire per chip 43B transistors	20% core processor area reduction 15% power reduction +11% single thread performance
Capacity	200 cores	208 cores	+15-20% capacity
Cache	32MB L2 256MB virtual L3 2GB virtual L4	36MB L2 360MB Virtual L3 2.8GB virtual L4	+40% cache growth
Total system memory	4U DDR4 DDIMM 40TB max memory	4U DDR5 DDIMM 64TB max memory	+60% system memory growth
On-chip AI acceleration	Telum	Telum II	More AI processing per chip Up to 8x on-chip AI processors available per core, 7.5X more throughput Support for LLM compute primitives Int8, FP16 datatypes; quantization for improved efficiency
Off-chip PCIe Accelerator card	N/A	Spyre AI accelerator chip 32 Gen AI-ready cores on extended adapters 75W PCIe gen5 x16 adapter Up to 48 adapters per system	Available only on IBM z17
System I/O	Off-chip IO processing	On-processor chip IO DPU	70% Reduced power for IO management Double density with new FICON Express 32G (4-port) Double density with converged network adapter (RoCE and OSA)

Hardware and system structure improvements

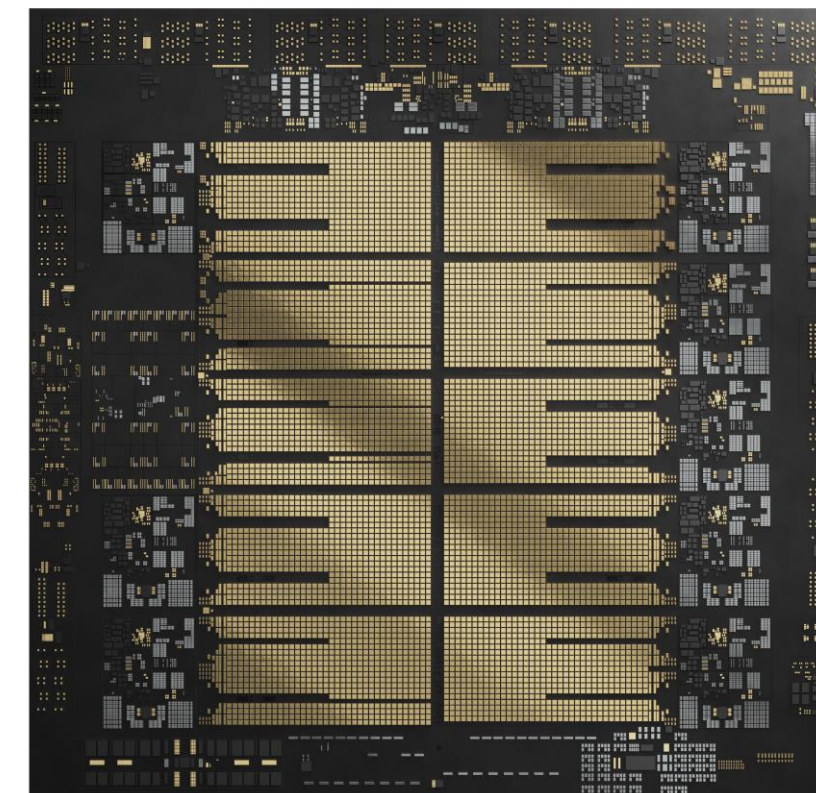
Comparison of IBM z15 to IBM z17

	z15	z17	z17 differentiation
Processor Chip	14nm technology 5.2GHz 12 cores per chip 14.5 miles of wire per chip 9.2B transistors	5nm technology 5.5 GHz 8 cores per chip 24.1 Miles of wire per chip 43B transistors	Core processor instruction set and performance improvements +21% single thread performance
Capacity	190 cores	208 cores	+35% capacity
Cache	4MB L2, 256MB L3, 960MB L4	36MB L2, 360MB Virtual L3, 2.8GB virtual L4	+200% total cache growth
Total system memory	40TB max memory	64TB max memory 4U DDR5 DDIMM	+60% system memory growth
On-chip AI acceleration	No AI acceleration	Telum II 2 nd -gen on-chip AI accelerator	On-processor ultra-low latency AI inference engine Up to 8x on-chip AI processors available per core Support for LLM compute primitives Improved quantization: Int8, FP16 datatypes Multi-model real-time AI on 100% of transactions
Off-chip PCIe Accelerator card	Not available	Spyre AI accelerator card 32 Gen AI-ready cores Up to 48 PCIe gen5 x16 adapter	Available only on IBM z17 On-prem watsonx Code Assistant and watsonx Assistant Supported LLMs, generative, and agentic AI with Spyre
System I/O	Off-chip IO Processing	On-processor chip IO DPU	70% Reduced power for IO management Double density (Up to -21% system power for I/O rich configs) - FICON Express 32G (4-port) - Converged network adapter (RoCE and OSA)
Other features	Not available	Quantum-Safe Secure Boot Memory Encryption Flexible Capacity for Cyber Resiliency	Significantly improved security and compliance Infrastructure and compute agility Consumption-based model for flexible compute

Predictive AI and **Encoder LLMs**
for real-time, high-volume
transaction processing
unlocking business growth

Decoder LLMs to accelerate and
scale **Generative AI** workloads

IBM Telum II™



IBM Spyre™ Accelerator



Why IBM AI Spyre Accelerator in IBM z17

1. LLMs and generative AI on IBM z17

- Deploy encoder LLMs
- Deploy generative AI applications like WXC4Az and WXA4z
- Deploy watsonx.ai

2. Full stack support on IBM Z

- Maximum convergence of AI software, compilers, open source and hardware
- Drive technical and organizational alignment across all layers of the stack

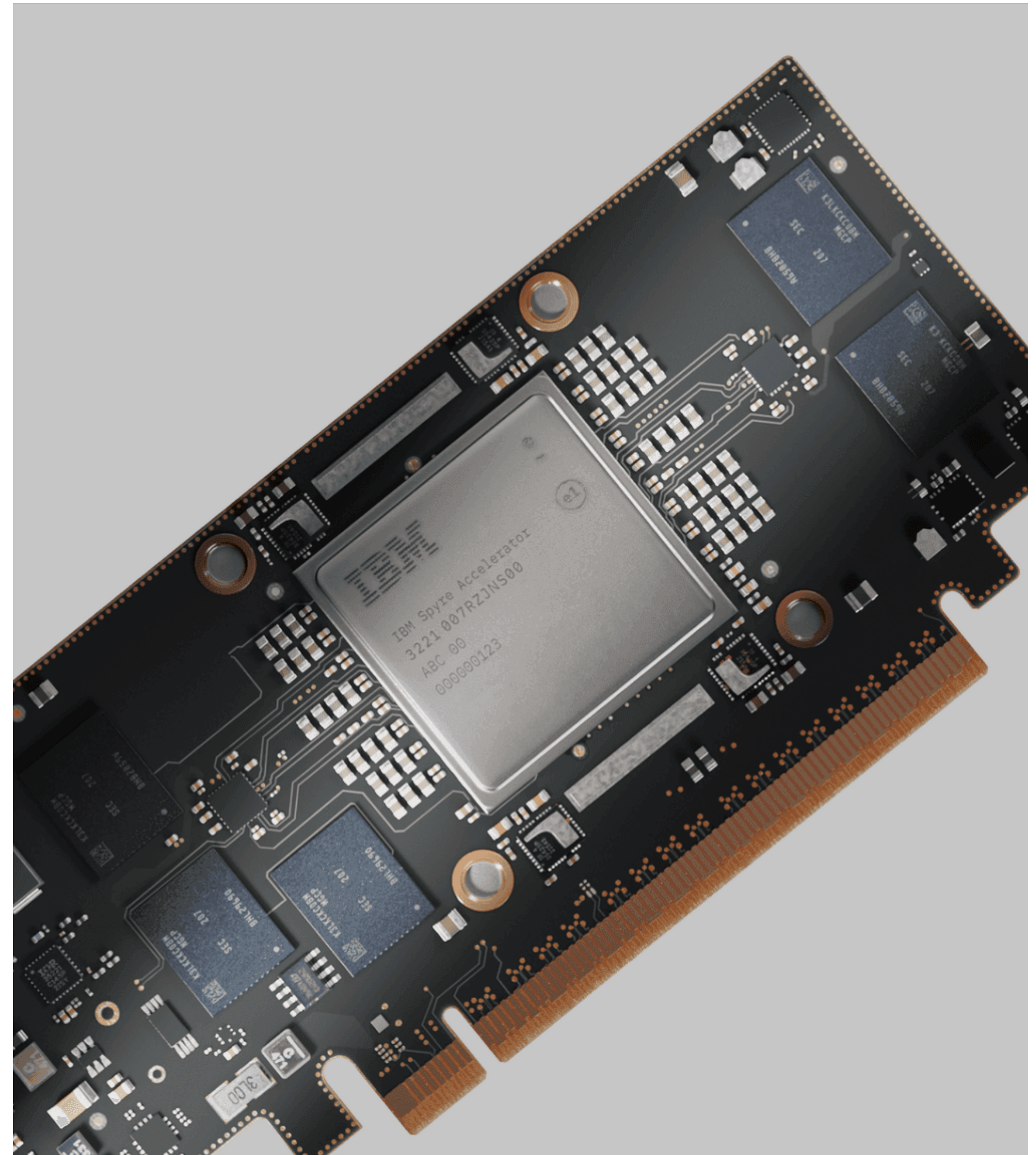
3. Resiliency, Availability, Serviceability & Security

- Scalable architecture ensuring highly available deployments
- Protect data and application logic
- IBM Z® qualities of service

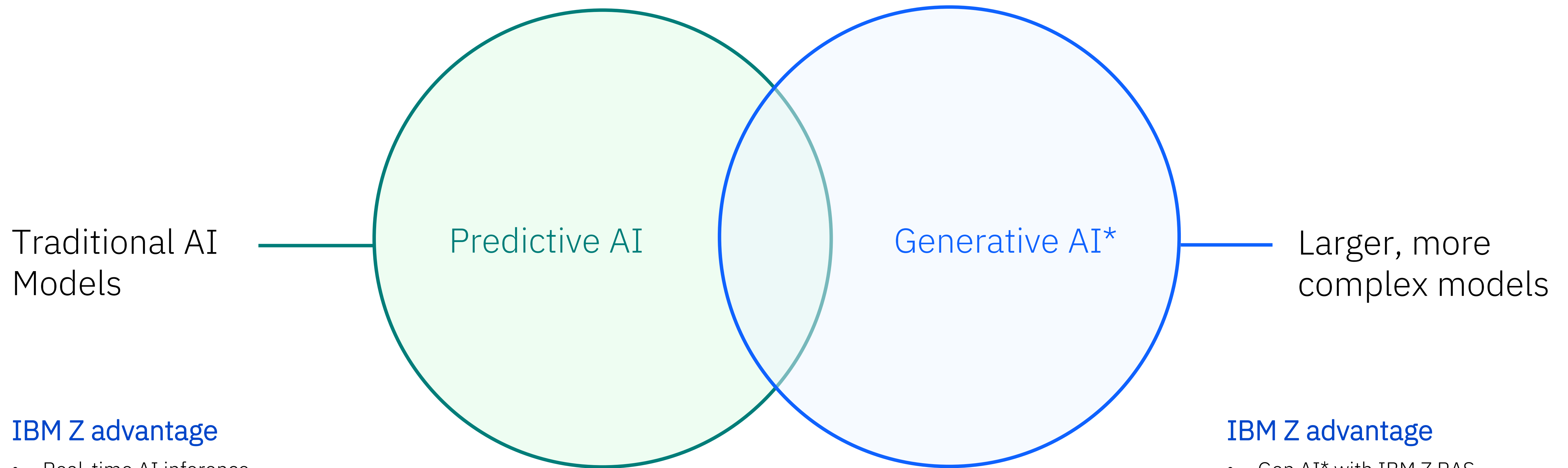
4. Invest for future AI growth on IBM Z*

- Agentic AI for AIOps and system optimization use cases
- Leverage commercial proprietary and opensource LLMs
- Card virtualization and fine-tuning of LLMs

*Futures statement subject to change



Traditional AI and Gen AI on IBM Z deliver unique capabilities



IBM Z advantage

- Real-time AI inference
- In-transaction AI at scale
- Inference on every transaction¹

IBM Z advantage

- Gen AI* with IBM Z RAS
- Security and data protection
- Energy efficiency

Real-time multiple-model AI only on IBM z17

Generates high value by combining the power of traditional AI models and LLMs in real-time on every transaction which can lead to higher predictive accuracy and expanded insights

* With IBM Spyre AI Accelerator

Statement of Direction - IBM Spyre AI Accelerator

IBM Spyre AI Accelerator: IBM intends to deliver the IBM Spyre AI Accelerator on Z, which is a PCIe-attached AI accelerator designed to support products, assistants, and services that leverage large language models on the IBM Z platform. The IBM Spyre AI Accelerator on IBM z17 is designed to bring generative AI to mainframes, empowering enterprises to scale-up AI capabilities and workloads. Clients will be able to deploy large language models (LLMs) including encoder, decoder, encoder-decoder architecture LLMs on the IBM Z platform, and benefit from the robust security, performance, and reliability it offers. IBM Z software products that will depend on Spyre AI Accelerators for full on-prem deployment include watsonx Code Assistant for IBM Z and watsonx Assistant for IBM Z. A hybrid deployment model is available for watsonx Assistant for IBM Z before Spyre AI Accelerator availability, as [announced here](#).

Other AI on IBM Z products that are also planned for Spyre AI Accelerator exploitation upon availability include Machine Learning for IBM z/OS, and components of AI Toolkit for IBM Z & LinuxONE like IBM Z Accelerated for Nvidia Triton Inference Server and IBM Z Accelerated for PyTorch. IBM Spyre AI Accelerator is planned to be available starting in 4Q 2025, in accordance with applicable import/export regulations.

Marketing, segmentation,
and client readiness

IBM z17

Market and goals

Communications& Marketing goals

Increase positive awareness and perception of brand

Increase the presence of the client voice

Improve audience understanding of key themes to elevate full-stack value

Reach targets with positive media, influencer and analyst content

Audiences

Industries:
Banking / FSS, Insurance, Technology, Government

Primary Personas:
CIO, CTO, Dir of IT, VP of Infra, CDO, CISO:

Secondary Personas
LOB Leads, Engineering Leads, IT Mgr/Architects, App Leads, Data Scientists

Segmentation

Progressive – accounts that show a propensity to refresh every generation

Generation Skipper – accounts that skip the current generation for the next

At Risk – accounts that wait at least 2 generations to update

Cluster breakdown

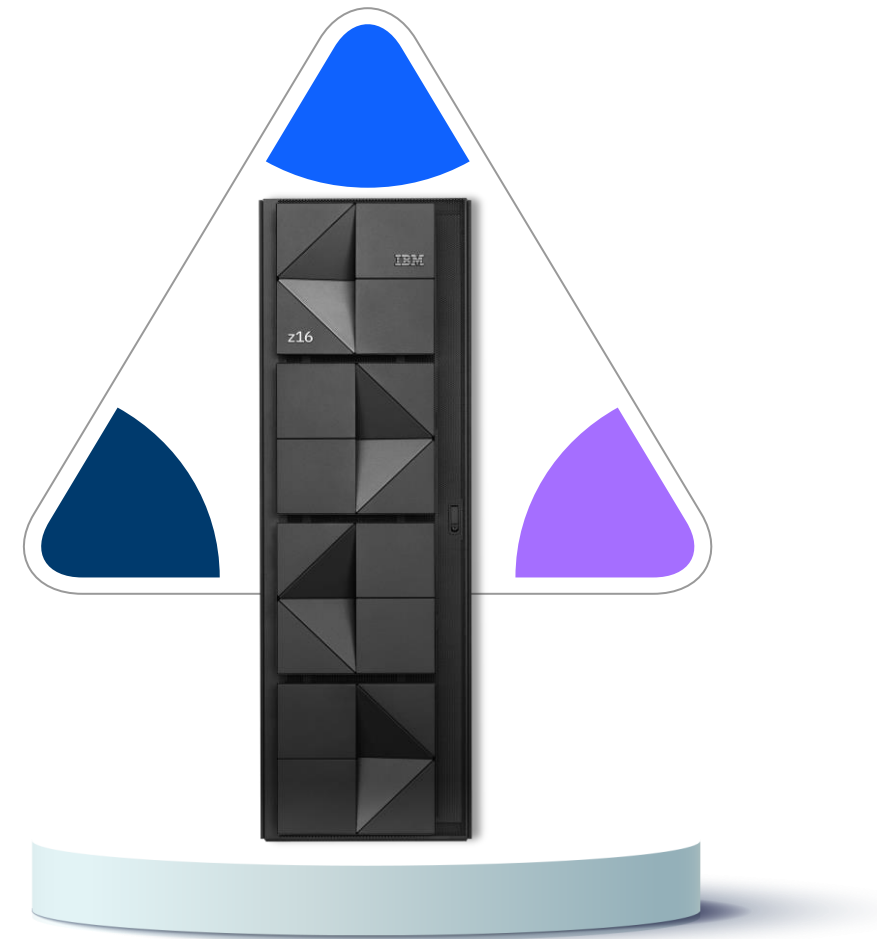
Marketing by buying behavior

Cluster	Business Goal	Marketing Goal	Roles	Sales/Marketing Led
Progressive	Shorten deal cycles – Save money now vs. later	Create a sense of urgency, marketing AI through the stack promoting automation’s cost effectiveness as a key business value	CISO, CIO, Director of IT, VP of Infra	Sales
Skipper	Get out of generation skipping mentality, but also happy to keep them in the family	Key to focus on promoting business value only (not as financially sensitive) emphasizing responsibility for business growth	Business messaging – VP of Infrastructure, CISO, Chief Data Officer; Technical messaging – Enterprise Architect, LOB, CIO	Primarily Marketing + BP Ecosystem
At Risk	Retain and expand the Z family	Highlight key product points to modernize and upgrade SW + HW	Business leaders	Marketing
Generic	Awareness and perception: Increase the visible presence of the customer’s voice	Increase audience understanding of key concepts to elevate full stack value proposition. Value of AI Acceleration, multi-model AI, AI-enabled security, etc. Account segmentations: All Z install base	Dir of IT, VP of Infra, CIO, CDO, CISO + Influencers (LOB leads, engineering leads, IT Mgr/Architects, App Leads, Data Sciences)	Marketing

IBM z17 client readiness & pre-conditioning

Prepare clients for next generation of zHW

- Ensure clients are meeting all pre-requisites to migrate to IBM z17
- Leverage the zStack to drive IBM z17 market adoption
 - Leverage Application Modernization (WCA4z, Db2v13)
 - Simplification and Automation (WXA4z, ITOps, Terraform, Ansible)
 - Exploit innovation by embedding multi-model AI (AI/Telum II/Spyre, Security, Resilience)
- **Pre-Condition clients with PoXs** on the topics we want to include in all sales proposals



Leading value-add Capabilities

Platform Simplification leveraging
genAI + Automation
– *watsonx Assistant for Z*

Application Modernization +
Code Optimization
– *watsonx Code Assistant for Z*

Enterprise AI/ML
embedded applications
– *zAI Platforms + zAI Tools*

Data Modernization
– *DB2 v13 ++ add-ons*

Enterprise IT Operations Mgmt &
Infrastructure Health
– *Intellimagic, Concert, wxa4z*

Business Continuity +
Cyber Resiliency
– *GDPS, CyberVault*

Resources

- [IBM z17 Sales Kit](#)
- [IBM z17 homepage](#) on ibm.com
- [IBM z17 \(9175\) Technical Guide](#) (Redbook)
- [IBM Z Global Skills Accelerator](#) Program
- 3D Interactive tour/demo (Kaon demo) for IBM [z17](#)
- IBM Technology Lifecycle Services for [IBM Z and IBM LinuxOne](#)
- Future-Proof Your IBM Z Infrastructure with [IBM Technology Lifecycle Services](#)

